

TASK-1

IP Class	IP Address Range	Default Mask	Subnet	Real-World Example
Class A	1.0.0.0 126.255.255.255	– 255.0.0.0		Large company or ISP network
Class B	128.0.0.0 191.255.255.255	– 255.255.0.0		University or college network
Class C	192.0.0.0 223.255.255.255	– 255.255.255.0		Home WiFi router or small office
Class D	224.0.0.0 239.255.255.255	– N/A		Live video streaming (e.g., IPTV)
Class E	240.0.0.0 255.255.255.255	– N/A		Research and experimental purposes

TASK-2

Given:

- **IP Address:** 192.168.10.45
- **Subnet Mask:** 255.255.255.0 (/24)

Step 1: Find the Network Address

A **/24** subnet means the first **24 bits** are the network part and the last **8 bits** are the host part.

- Network Address = **192.168.10.0**

Step 2: Find the Broadcast Address

Set all host bits to 1.

- Broadcast Address = **192.168.10.255**

Step 3: Find the Valid Host Range

- **First Host:** 192.168.10.1
- **Last Host:** 192.168.10.254

Final Answer

Item	Address
IP Address	192.168.10.45
Subnet Mask	255.255.255.0 (/24)
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
Valid Host Range	192.168.10.1 – 192.168.10.254

TASK-3

Subnetting Plan for a Food Delivery Startup (Using 10.0.0.0/24)

Group	Network Address	Subnet Mask	CIDR Host Range
Delivery Partners	10.0.0.0	255.255.255.192 /26	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64	255.255.255.192 /26	10.0.0.65 – 10.0.0.126

Group	Network Address	Subnet Mask	CIDR Host Range	
Customers	10.0.0.128	255.255.255.192 /26	10.0.0.129 10.0.0.190	—
Admins	10.0.0.192	255.255.255.192 /26	10.0.0.193 10.0.0.254	—

Explanation

- The **10.0.0.0/24** network is divided into **4 equal subnets**.
- Each subnet uses a **/26** subnet mask (**255.255.255.192**).
- Each subnet provides **62 usable host IP addresses**, which is suitable for each department.

TASK-4

Given

- **Network:** 172.16.0.0/16
- **Required Subnets:** At least **20**

Step 1: Find the Number of Bits Required

Use the formula:

$$[2^n \geq 20]$$

- $(2^4 = 16)$ ❌ (Not enough)
- $(2^5 = 32)$ ✅ (Enough)

So, **5 bits** are borrowed from the host portion.

Step 2: Calculate the New Subnet Mask

- Original prefix = **/16**
- Borrowed bits = **5**

New Prefix = /21

Subnet Mask = 255.255.248.0

Step 3: Calculate Hosts per Subnet

Remaining host bits:

$$[\begin{array}{rclcl} 32 & - & 21 & = & 11 \\ & & & & \end{array}]$$

Usable hosts:

$$[\begin{array}{rclcl} 2^{\{11\}} & - & 2 & = & 2048 & - & 2 & = & 2046 \\ & & & & \end{array}]$$

Final Answer

Item	Result
Original Network	172.16.0.0/16
Required Subnets	20
Bits Borrowed	5
New Prefix	/21
New Subnet Mask	255.255.248.0
Total Subnets	32
Usable Hosts per Subnet	2046

TASK-5

Answer

The IP address **192.168.1.300** is **invalid** because each part (octet) of an IPv4 address must be between **0 and 255**. Since **300** is greater than **255**, the device cannot connect to the network.

A valid IP address in the **192.168.1.0/24** subnet would be something like:

- **192.168.1.10**
- **192.168.1.50**
- **192.168.1.100**

The valid host range for this subnet is **192.168.1.1 to 192.168.1.254** (assuming **192.168.1.0** is the network address and **192.168.1.255** is the broadcast address).